

Name- Bhumika Shrotriya

Course Name – Data Analytics With GenAI

Subject – PowerBI

Topic – Modeling Data Assignment

Question 1: What is Data Loading in Power BI and why is it considered the first step of analysis?

Answer 1:

Data Loading in Power BI is the process of importing data from one or more external sources (such as CSV files, Excel sheets, SQL databases, APIs, etc.) into the Power BI data model, making it available for transformation, modeling, and visualization.

It is considered the **first step of analysis** because:

- Without data, no analysis, calculation, or visualization is possible. Everything in Power BI depends on having data loaded into the model.
- It establishes the foundation - column names, data types, and row-level data - on which all subsequent steps (cleaning, modeling, DAX, reports) are built.
- Power BI cannot detect relationships, suggest visualizations, or apply transformations until the data is brought in.
- It also defines the refresh strategy - whether data is static (one-time import) or dynamic (scheduled refresh from a live source).

Question 2: Explain the difference between “Load” and “Transform Data” in Power BI.

Answer 2:

Aspect	Load	Transform Data
What it does	Directly imports data into the Power BI data model as-is	Opens Power Query Editor before loading, allowing you to clean and reshape the data
When to use	When data is already clean and ready for use	When data needs cleaning (nulls, duplicates, type changes, splitting columns, etc.)
Power Query	Skipped — data bypasses Power Query Editor	Power Query Editor is opened where M-language steps are applied
Flexibility	Low — you get the data exactly as it exists in the source	High — you can filter rows, rename columns, merge tables, add columns, change types, etc.
Result	Data is loaded immediately into the Fields pane	After applying transformations, you click "Close & Apply" to load the cleaned data

Question 3: What is a Fact Table and a Dimension Table? Give examples from the dataset.

Answer 3:

Fact Table contains quantitative, measurable data (numbers/metrics) that represent business events or transactions. It is the central table in a star schema and usually has foreign keys linking to dimension tables.

Dimension Table contains descriptive, categorical attributes that provide context to the facts. They answer the "who, what, where, when" questions.

From the Coronavirus 2020 Dataset:

The dataset as a single table can be split into:

- **Fact Table (CoronaFacts):** Would contain the measurable columns —
 - Confirmed_Cases, Recovered_Cases, Deaths, Active_Cases, Tests_Conducted, Hospitalized
 - Plus foreign keys: Date, Country, State
- **Dimension Table — DimLocation:** Would contain —
 - Country, State (descriptive location attributes)
- **Dimension Table — DimDate:** Would contain —
 - Date, Month, Year, Quarter (time-based attributes for time-intelligence analysis)
- **Dimension Table — DimVaccination:** Would contain —
 - Vaccination_Status (Yes / No / Unknown — categorical attribute)

Question 4: Why is Star Schema preferred over Snowflake Schema in Power BI?

Answer 4:

a. Star Schema has one central Fact Table directly connected to multiple Dimension Tables (one level deep - no further normalization of dimensions).

b. Snowflake Schema is an extension where Dimension Tables are further normalized into sub-dimension tables (multiple levels deep).

Power BI prefers Star Schema because:

1. **Faster Query Performance:** Star Schema requires fewer joins between tables. Power BI's in-memory engine (VertiPaq) is optimized for simple one-level relationships, making queries significantly faster.

2. **Simpler DAX:** Writing DAX measures is much simpler when dimensions are flat. Snowflake schemas require navigating multiple table hops, making DAX formulas more complex and error-prone.
3. **Easier Relationships:** Power BI's Model View works best with simple one-to-many relationships between fact and dimension tables. Snowflake's multi-level relationships can create ambiguous or circular relationship paths.
4. **Better Performance with VertiPaq:** Power BI compresses and stores data column-by-column in memory. Denormalized dimension tables (Star Schema) compress very efficiently because they have repeating categorical values.
5. **Easier for Report Builders:** Business users and report authors find it easier to navigate a simple Star Schema in the Fields pane — all descriptive fields are one click away.

Question 5: Identify and Remove Duplicate Records based on Date, Country, and State.

Answer 5:

Finding from the dataset: After checking the actual data, there are **0 duplicate records** based on the combination of Date, Country, and State. The dataset is already unique on these three columns.

However, the steps to identify and remove duplicates in Power Query are:

Step 1 — Open Power Query Editor (Home → Transform Data).

Step 2 — Select the Date column, then hold **Ctrl** and also select the Country and State columns (so all three are selected).

Step 3 — Right-click on any of the selected column headers → click "**Remove Duplicates**". This keeps only the first occurrence of each unique combination of Date + Country + State and removes all subsequent duplicates.

Alternative method using "Keep Duplicates" to first identify them:

- Select the three columns → Home tab → **Keep Rows** → **Keep Duplicates**. This shows you only the duplicate rows so you can inspect them before removing.

Question 6: Identify and Replace Null Values in Vaccination_Status.

Answer 6:

Finding from the dataset: The Vaccination_Status column has **170 null values** out of 500 rows (34% missing). The existing values are: **Yes (178), No (152), Null (170)**.

Steps in Power Query to replace nulls:

Step 1 - In Power Query Editor, click on the Vaccination_Status column header to select it.

Step 2 - Go to **Transform** tab → click **Replace Values**.

Step 3 - In the dialog:

- **Value to Find:** leave blank (or type null)
- **Replace With:** type Unknown
- Click **OK**

Step 4 - All 170 null values will now be replaced with "Unknown".

Alternative - Right-click method: Right-click the Vaccination_Status column → **Replace Values** → set null → "Unknown".

After replacement, Vaccination_Status will have:

- Yes → 178
- No → 152
- Unknown → 170 (previously null)

Question 7: Create a new column to calculate Recovery Rate.

Answer 7:

Recovery Rate = (Recovered Cases / Confirmed Cases) × 100

In Power Query (Custom Column):

Go to **Add Column** tab → **Custom Column** → Name it Recovery_Rate → Enter the formula:

= [Recovered_Cases] / [Confirmed_Cases] * 100

Question 8: Summarized Table Showing Total Confirmed Cases by Country.

Answer 8:

From the actual dataset, the summarized result is:

Country	Total Confirmed Cases
Brazil	4,804,531
Spain	4,354,613
India	4,198,680
UK	3,651,598
USA	3,399,318
Italy	3,390,760

How to create this in Power BI:

Method 1 — Using a Table Visual (Report View):

- Add a **Table** visual to the canvas.
- Drag Country into the **Columns/Values** field.
- Drag Confirmed_Cases into the **Values** field (it will auto-sum).
- Sort by Confirmed Cases descending.